
Hardware Side Channels

Secure Software and Hardware Systems

Goal of this Lecture

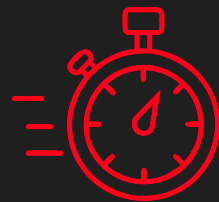
- Familiarize you with the concept of side-channel attacks
- Give some examples about side-channel defenses

Overview

- Introduction
- Power Side Channels
 - Idea & Example
 - Power Consumption Models
 - Types of Power Side Channel Analysis
- Defenses
- Recap

Motivation

```
bool check_pin(int pin_entered[4])
{
    for(int i = 0; i < 4; i++) {
        if(pin_stored[i] != pin_entered[i]) {
            return false;
        }
    }
    return true;
}
```



Motivation

0000		4000		4900		4910	
1000		4100		4910		4911	
2000		4200		4920		4912	
3000		4300		4930		4913	
4000		4400		4940		4914	
5000		4500		4950		4915	
6000		4600		4960		4916	
7000		4700		4970		4917	
8000		4800		4980		4918	
9000		4900		4990		4919	

```
bool check_pin (int pin_entered [4])
{
    for (int i = 0; i < 4; i++) {
        if (pin_stored[i] != pin_entered[i]) {
            return false;
        }
    }
    return true;
}
```



PIN = 4919

Worst case:
4*10 trials
instead of 10,000(!)

Definition: Side Channels

“A *side channel* is where information leaks accidentally via some medium that was not designed or intended for communication.”

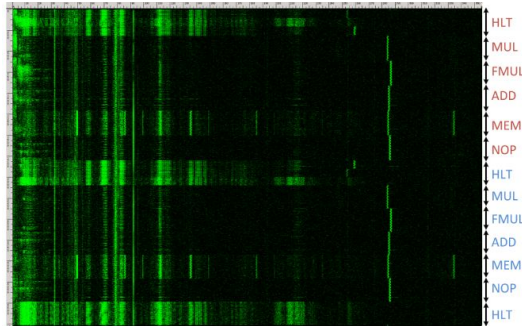
– Ross Anderson [1]



AS SIDE-CHANNEL ATTACKER, WE ABUSE THIS LEAKAGE TO EXFILTRATE DATA

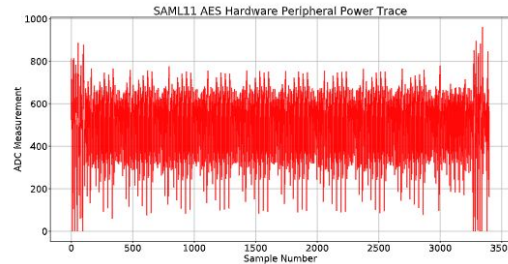
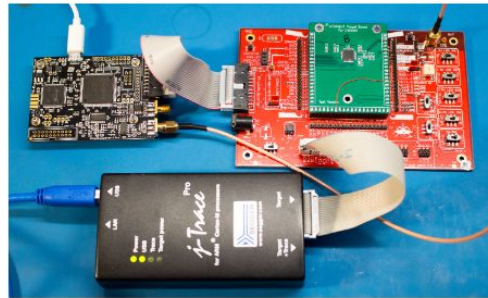
Side Channels: Examples

Acoustic



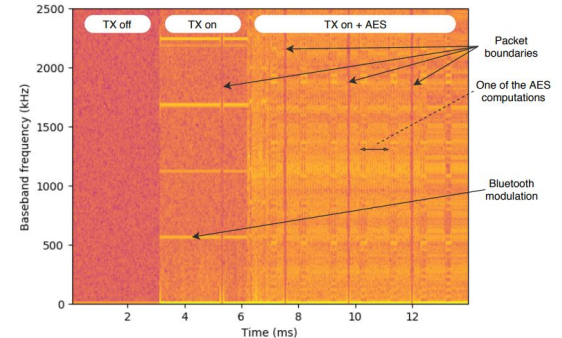
Genkin et al. "RSA Key Extraction via Low-Bandwidth Acoustic Cryptanalysis"

Power



O'Flynn & Dewar. "On-Device Power Analysis Across HardwareSecurity Domains. Stop Hitting Yourself."

Electro Magnetic



Camurati et al.. "Screaming channels: When electromagnetic side channels meet radio transceivers"

Overview

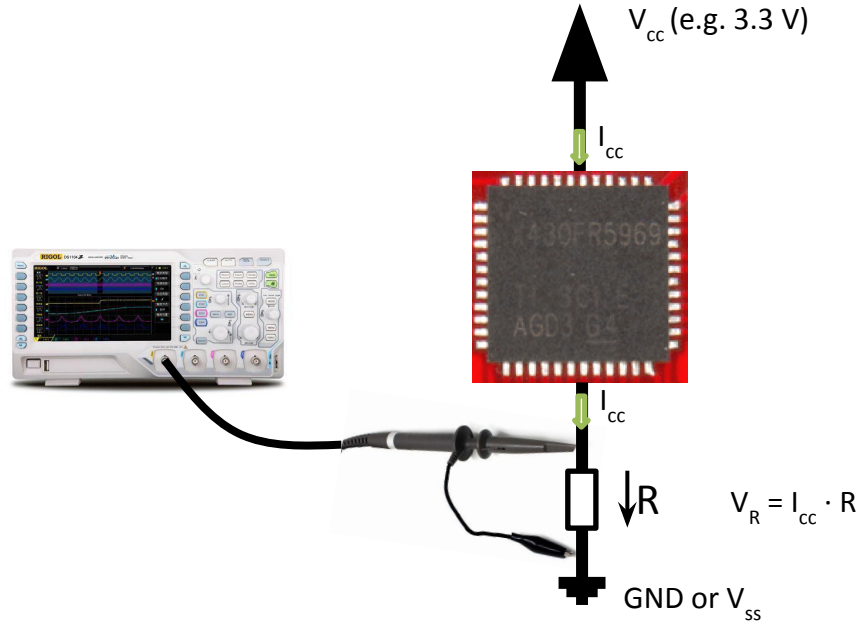
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Power Side Channels

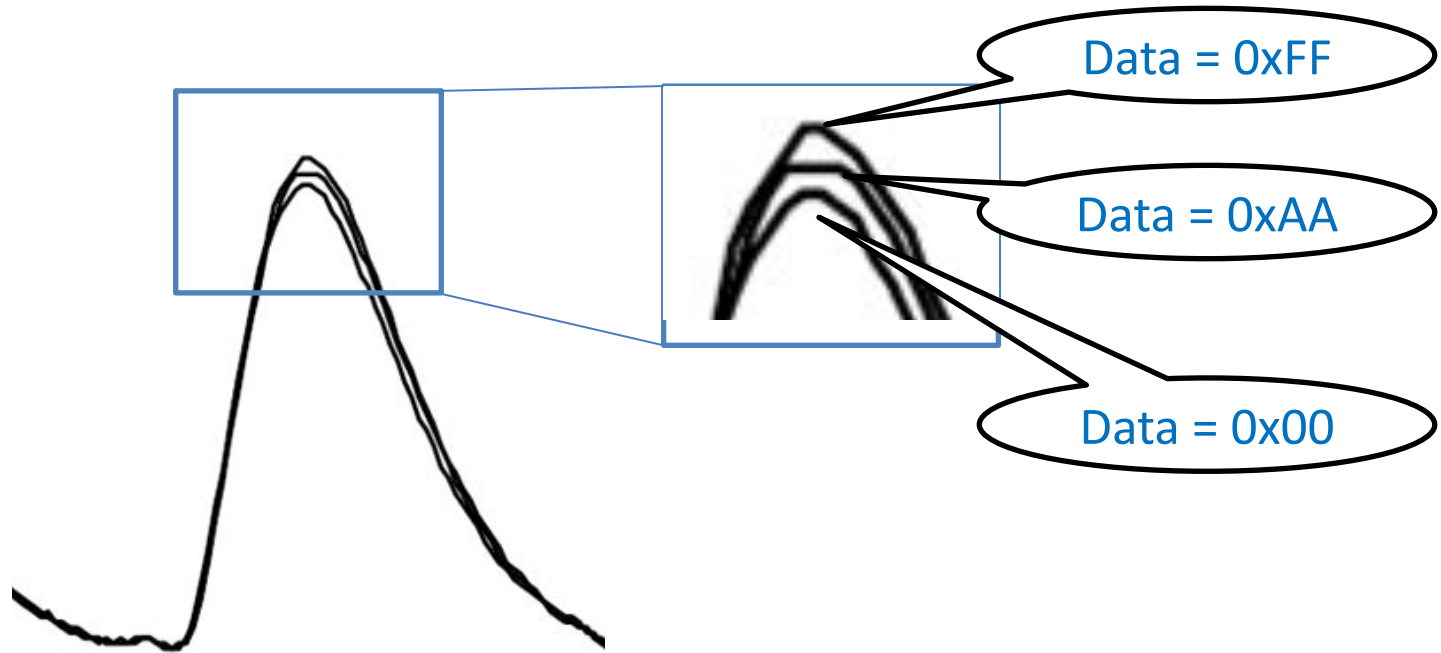
- Computation on device requires energy
 - We can inspect energy consumption!
- In the easiest case, we can differentiate between
 - Activity
 - No-Activity
 - Proxy for timing side-channels!
- Some operations may require more energy than others
 - 💡 This is another side channel!



Measuring Power Consumption

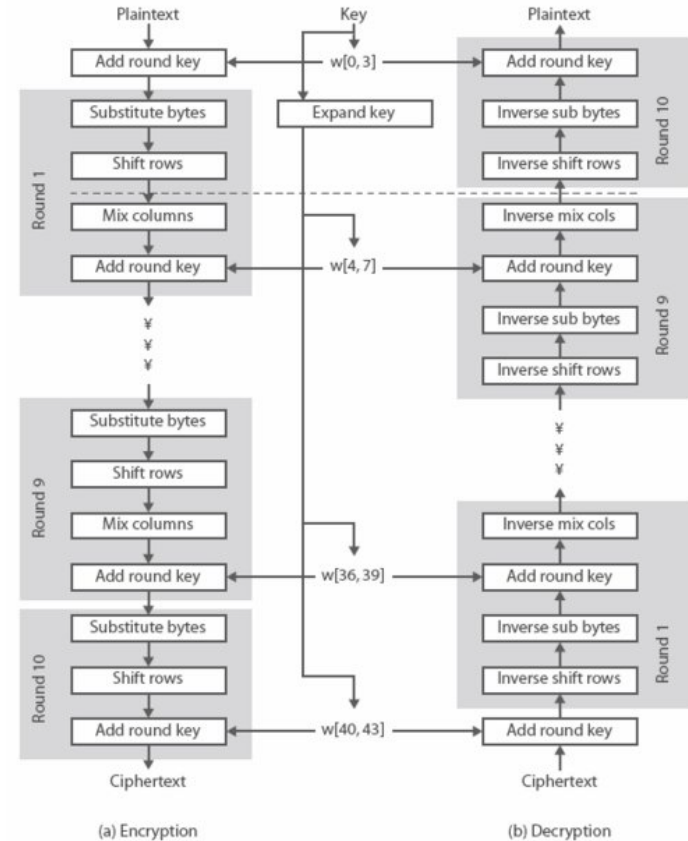


Power Consumption: Simplified Idea

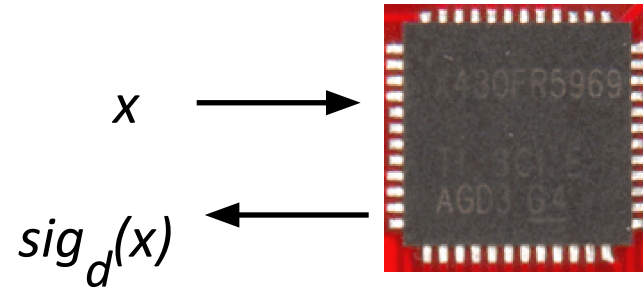


Live Demo

- AES: Symmetric Encryption
 - 10 Rounds
 - Different parts with different operations
- Can we see this in a power trace?



Example: RSA signature / decryption



Square-and-multiply

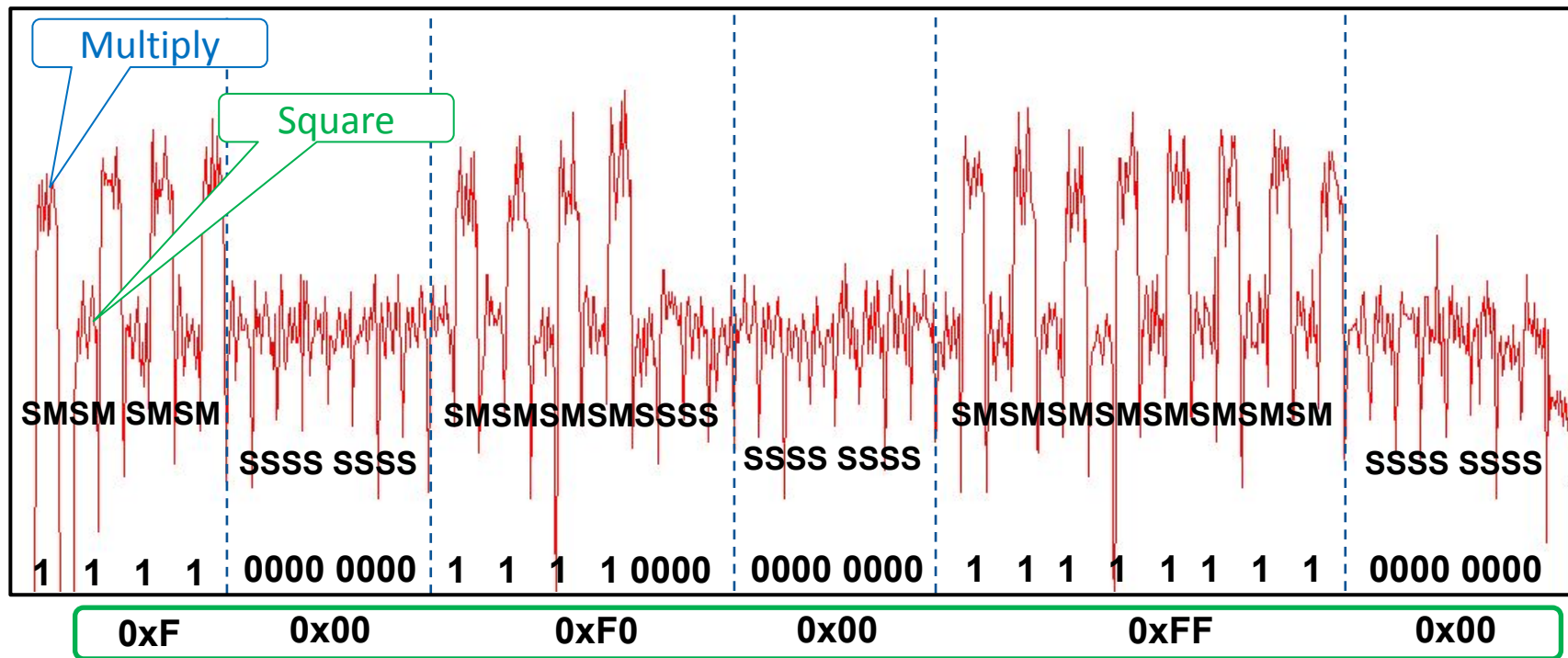
- Textbook RSA Signature:

$$s = m^d \bmod n$$

- Square and multiply: efficient exponentiation
 - Calculation dependent on individual key bits
 - $\text{key}[i] == 0 \rightarrow \text{Square}$
 - $\text{key}[i] == 1 \rightarrow \text{Multiply}$

```
s = 1
for i = 2048 - 1 ... 0
    s = s2 mod n
    if bit i of d == 1
        s = s * m mod n
    end if
end for
return s
```

Square & Multiply



key value : 0xF 00 F0 00 FF 00

Insight

Even if mathematical secure and sound, cryptographic algorithms may leak secrets via side-channel analysis!

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Power Consumption Models

To model leakage (as attacker), we can make an easy simplification:

- Power consumption depends on the bit values being processed.

Having a power consumption model, allows us to find potential points of attacks

- Sometimes also referred to as “Leakage Assumption”



Hamming Weight

- Hamming Weight is the numbers of bits set in a bit vector, e.g.,
 - $\text{HW}(0b10000110) = 3$
 - $\text{HW}(0b00000000) = 0$
 - $\text{HW}(0b11111111) = 8$
- Core intuition: A bit set requires more energy in hardware
 - Does not hold true in practice!
 - Still, good enough for side-channel analysis



Hamming Distance

- Hamming Distance is the numbers of bits changed between two bit vectors, e.g.,:
 - $HD(0b00001111 \rightarrow 0b00000011) = 2$
 - $HD(0b00000000 \rightarrow 0b11110000) = 4$
 - $HD(0b00000000 \rightarrow 0b11111111) = 8$
- Core intuition: Toggling bits requires energy in hardware
 - Closer to reality, yet still not entirely correct
 - Very good results in practice



Overcoming Noise

- A key issue complicating side-channel analysis is noise
 - Other operations on the system
 - Environmental interference
 - Measurement setup
- Multiple possible solutions (which may require many traces):
 - Filtering/Signal Processing
 - Averaging
 - Statistical Analysis
 - Machine Learning

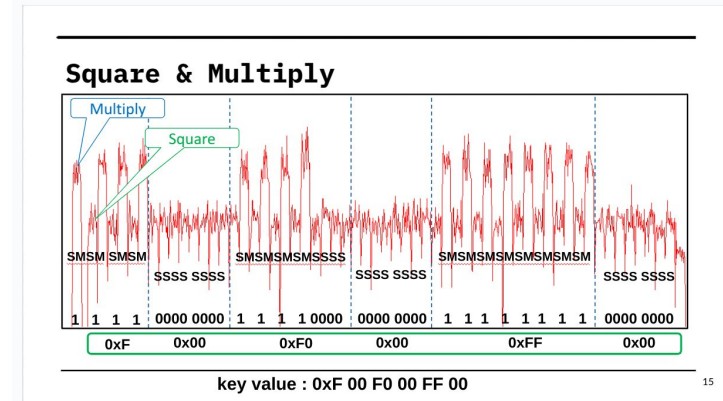


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Simple Power Analysis (SPA)

- Inspect trace(s) of a single operation
 - May even be based on manual inspection
- Requirement:
 - Different operations of system clearly distinguishable
- Multiple traces may help to reduce noise



Differential Power Analysis (DPA)

- Requires large amount of traces
 - Collected under different executions of same algorithm
- Strategy:
 - Iterate over different (known) plain text bytes
 - Place a (sub)key guess k_{guess}
 - Group traces into assuming k_{guess} is correct
 - Calculate *difference of means* within groups
 - If k_{guess} is correct: visible peak
- Exploits data-dependence of computation
 - Useful if program flow is the same regardless of data
 - Ties directly to power consumption modules
 - Is pretty effective in presence of noise

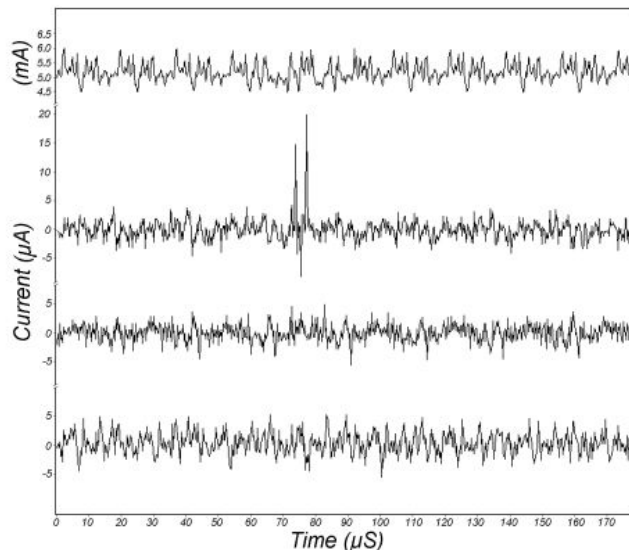


Figure 4: DPA traces, one correct and two incorrect, with power reference.

Correlation Power Analysis (CPA)

- Different approach to DPA
 - Requires less traces than difference-of-means approach
- Strategy:
 - Decide on power model
 - Apply model to critical point in algorithm
 - Model power consumption for different values under k_{guess}
 - Measure actual power consumption of device
 - Perform correlation analysis between hypothetical and actual power consumption
 - If guess is correct, high correlation visible

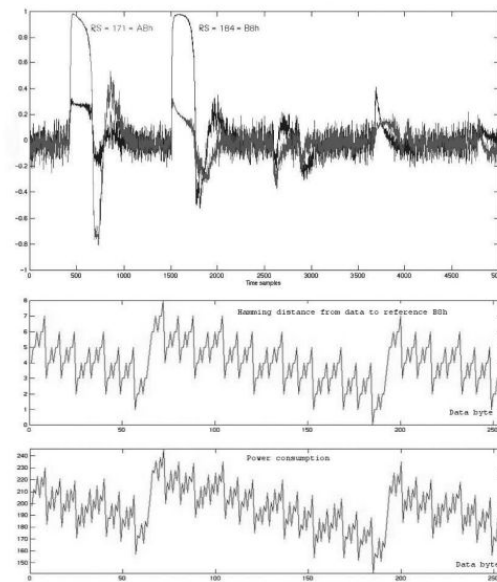


Fig. 1. Upper: consecutive correlation peaks for two different reference states. Lower: for varying data (0-255), model array and measurement array taken at the time of the second correlation peak.

Template Attacks

- Core idea: Use power consumption templates
 - Assumes that same device behaves deterministic
 - Requires very small amount of trace
- Strategy:
 - Run algorithm on target hardware with known keys and create templates
 - Match measurements to the created templates
- Effective, but requires:
 - Target knowledge
 - Time for profiling

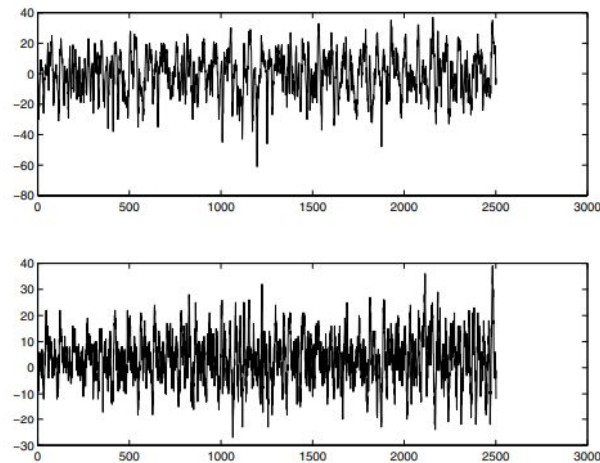


Fig. 1. Differences of side channel samples: upper figure is for the same key while the lower is for two different keys

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Masking

- Idea: randomly split every sensitive variable into $d + 1$ shares
 - d : Order of the masking
- Sensitive variable is never directly accessed
 - All computations are done on the shares
- Attacker now needs to correctly determine all shares
 - Thwarts attacks if masks are regenerated over different runs in a leak free environment [1]



Blinding

- Adding entropy via random values to the computation
 - So that the operation interesting for SCA does not operate on the sensitive value
- Strategy:
 - Before cryptographic operation: mix-in random value
 - Perform cryptographic operation
 - After cryptographic operation: remove blinding factor



Blinding Example: RSA

Defense against *timing* side-channel revealing m during decryption

Blinding: $z = mr \bmod N$

Encryption: $c = z^e \bmod N$

Decryption: $z = c^d \bmod N$

Unblinding: $m = zr^{-1} \bmod N$

e : public key

d : private key

N : modulus

r : Blinding factor, relative prime to N

c : ciphertext

m : message

z : blinded message

ATTACKER DOES NOT KNOW R , SIDE-CHANNEL
MEASUREMENT WILL NOT YIELD REAL M



Cryptographic Accelerators

- Idea: Use special co-processors to perform sensitive operations
 - Implementation of cryptographic algorithms in hardware
- Usually hardened against side-channel analysis
 - Analysis not impossible, but often more difficult
- Examples:
 - AES-NI (Intel & AMD CPUs)
 - AES/DES on CRYPP co-processor for STM32MP1



Constant Time Operation

- Ensure that algorithm run time is not dependent on data/inputs
 - Main defense against timing attacks
- Requirements:
 - No data-dependent branches
 - No data-dependent memory accesses

← *MORE LATER!*

Constant Time: Example

```
bool check_pin(int pin_entered[4])
{
    for(int i = 0; i < 4; i++) {
        if(pin_stored[i] != pin_entered[i]) {
            return false;
        }
    }
    return true;
}
```



```
bool check_pin(int pin_entered[4])
{
    int ret = 0
    for(int i = 0; i < 4; i++) {
        ret |= pin_stored[i] ^ pin_entered[i];
    }
    return (ret == 0);
}
```

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Recap

- Looked into general side-channel
 - “What” are side channels?
 - “Why” are they a problem?
- Investigated power side-channels
 - “How” do some of the fundamental SCA approaches work?
- Discussed defense strategy
 - “How” can we mitigate side-channel attacks?

Where to learn more?

- The different papers linked in this lecture
- Power Analysis attacks: revealing the secrets of Smart Cards (Mangard, Oswald & Popp)
- HWSec course by Renault Pacalet:
 - <https://perso.telecom-paristech.fr/pacalet/HWSec/>
- The Hardware Hacking Handbook (Woudenberg & O'Flynn)
- ChipWhisperer Platform:
 - <http://wiki.newae.com/>
 - Also good for learning about FI!

